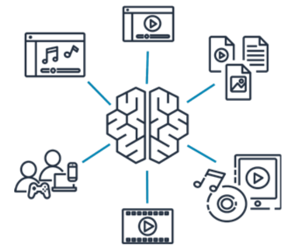
Exploring Artificial Intelligence Use Cases and Applications

Introduction

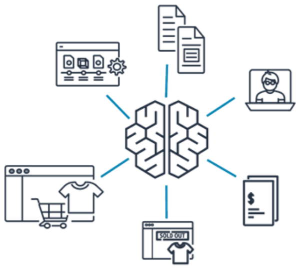
Explore real-world use cases in artificial intelligence (AI), machine learning (ML), and generative artificial intelligence (generative AI) across a range of industries. These areas include healthcare, finance, marketing, entertainment, and more.

Examples of Real-World Use Cases



Media and Entertainment

- **Content generation:** AI can create scripts, dialogues, or even complete stories for films, TV shows, and games.
- **Virtual reality:** AI can create immersive and interactive virtual environments for games or simulations.
- **New generation:** AI can generate articles or summaries based on raw data or events**.**



Retail

- **Product review summaries:** AI can generate review summaries for products so consumers can quickly find pertinent information.
- **Pricing optimization:** AI can model different pricing scenarios to determine optimal pricing strategies that maximize profits.
- **Virtual try-ons:** AI can generate virtual models of customers for virtual try-ons, which can improve the online shopping experience.
- **Store layout optimization:** AI can generate the most efficient store layouts to improve the customer shopping experience and boost sales.

Healthcare

- **AWS HealthScribe:** This AWS service empowers healthcare software vendors to build clinical applications that automatically generate clinical notes by analyzing patient-clinician conversations.
- **Personalize medicine:** By generating treatment plans based on a patient's specific genetic makeup, lifestyle, and disease progression, AI can contribute to more effective, personalized care.
- **Improve medical imaging:** AI can enhance, reconstruct, or even generate medical images, like X-rays, MRIs, or CT scans, which can aid in better diagnosis.

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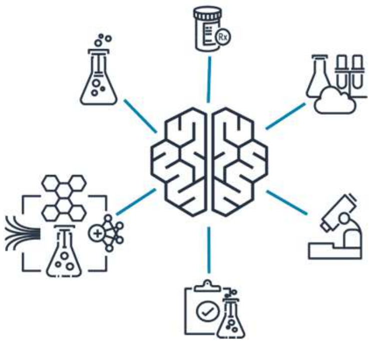
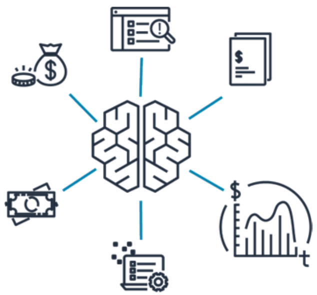


- **Life Sciences**
  - **Drug discovery:** AI can generate new potential molecular structures for drugs and accelerate the process of drug discovery and reducing costs.
  - **Protein folding prediction:** AI can predict the 3D structures of proteins based on their amino acid sequence, which is crucial for understanding diseases and developing new therapies.
  - **Synthetic biology:** AI can generate designs for synthetic biological systems, such as engineered organisms or biological circuits.



Financial Services

- **Fraud detection mechanisms:** AI can help create synthetic datasets to improve AI and ML systems by simulating various money-laundering patterns.
- **Portfolio management:** AI can simulate various market scenarios and help in the creation and management of robust investment portfolios.
- **Debt collection:** AI can generate the most effective communication and negotiation strategies for debt collection to increase the rate of successful collections.

Examples of Real-World Use Cases

Manufacturing

- **Predictive maintenance:** By analyzing historical production data, AI can predict maintenance schedules that will provide the most efficient machine outputs and reduce downtimes.
- **Product design:** AI can be used to create new product designs based on set parameters and constraints. It can generate multiple design options and optimize for factors like cost, materials, performance, and so forth.
- **Material science:** AI can help generate new material compositions with desired properties.

Examples of AI Applications - Computer Vision

- **Autonomous Driving** Auto manufacturers can use computer vision technology to make self-driving cars safer and more reliable.

Business value: Enhance customer experience

- **Healthcare/Medical Imaging** Using computer vision in healthcare can improve the accuracy and speed of medical diagnoses, which leads to better treatment outcomes and increased life expectancy for patients.

Business value: Improve business operations

- **Public Safety and Home Security** Computer vision image and facial recognition can swiftly identify unlawful entries or persons of interest, which fosters safer communities and works as a crime deterrent.

Business value: Enhance customer experience

Examples of AI Applications – Natural Language Processing

- **Insurance** companies can use NLP to extract policy numbers, expiration dates, and other personal information.

Business value: Sensitive data redaction

- **Telecommunication** companies use NLP to analyze customer text messages and suggest personalized recommendations.

Business value: Customer engagement

- **Education** In the education industry, students use Q&A chatbots to address questions.

Business value: Enhance student experience and engagement

Examples of AI Applications – Intelligent Document Processing

- **Financial services** use IDP to extract important information from mortgage applications to accelerate customer response time. It also helps with the underwriting process by identifying incomplete loan packages, tax forms, pay stubs, and other missing data.

Business value: Improve business operations, automation

- **IDP**, along with other applications such as optical character recognition (OCR) and NLP, helps eliminate the manual effort of processing documents such as contractual documents, agreements, court filings, and legal dockets.

Business value: Improve business operations

- Using IDP in **healthcare** can help expedite business quickly and accurately by processing various document types, such as claims and doctor's notes.

Business value: Improve business operations

Examples of AI Applications – Fraud Detection

- **Financial services** use fraud detection for identity verification, payment fraud detection, transaction surveillance, and anti-money laundering (AML) sanctions.

Business value: Improve business operations

- **Retail** Fraud detection systems in the retail industry protect businesses from financial losses, safeguard customer accounts and data, and maintain trust and confidence in online transactions.

Business value: Improve business operations

- **Telecommunication** industry uses fraud detection to identify any fraudulent activities in any of the following areas:
  - **Telecom** Roaming, premium rate service, and subscription fraud
  - **Online** New account fraud, claims processing fraud, and promotion abuse
  - **Retail** Credit card and online retail fraud

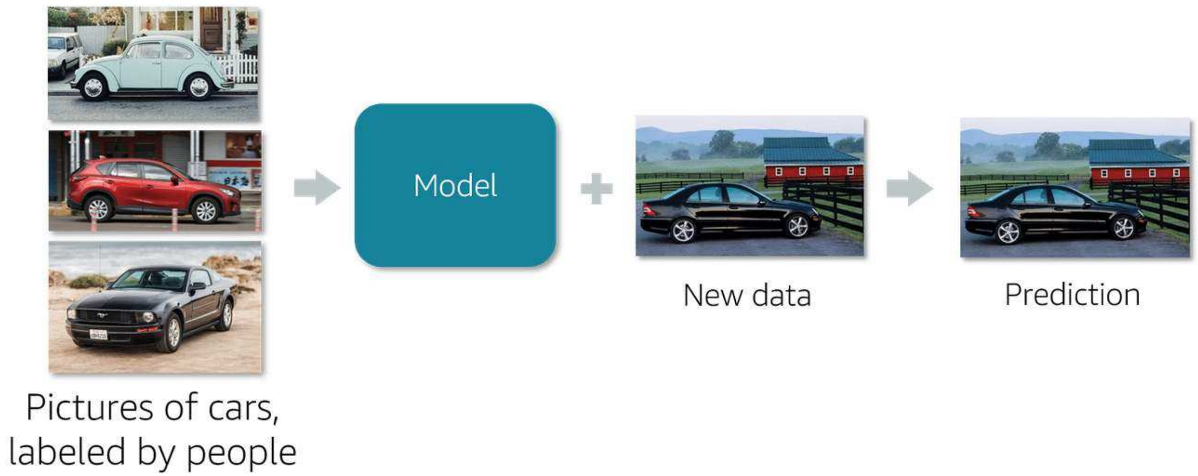
Business value: Improve business operations

Machine Learning Techniques and Use Cases

ML techniques

- Supervised learning
- Unsupervised learning
- Reinforcement learning

- Supervised learning uses labeled training data (the "supervisor") to teach algorithms through examples.
- The model learns patterns and relationships between inputs and outputs from this labeled data.
- Once trained, it can make predictions on new, unlabeled data based on what it has learned.



Types of Supervised ML

- **Regression** is a supervised learning technique used for predicting continuous or numerical values based on one or more input variable.

Use cases include the following:
- Advertising popularity prediction
- Weather forecasting
- Market forecasting
- Estimating life expectancy
- Population growth prediction

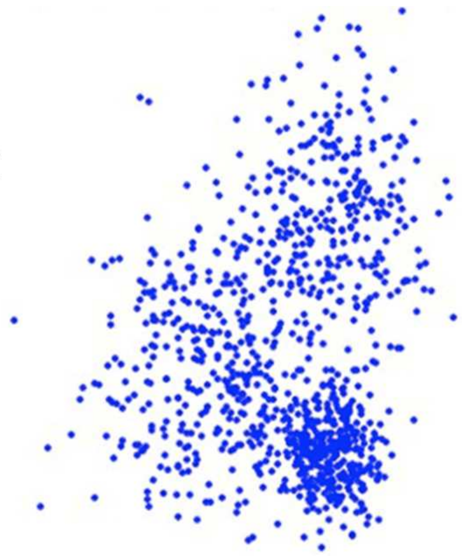
- **Classification** is a supervised learning technique used to assign labels or categories to new, unseen data instances based on a trained model.

- The model is trained on a labeled dataset, where each instance is already assigned to a known class or category.

Use cases include the following:
- Fraud detection
- Image classification
- Customer retention
- Diagnostics

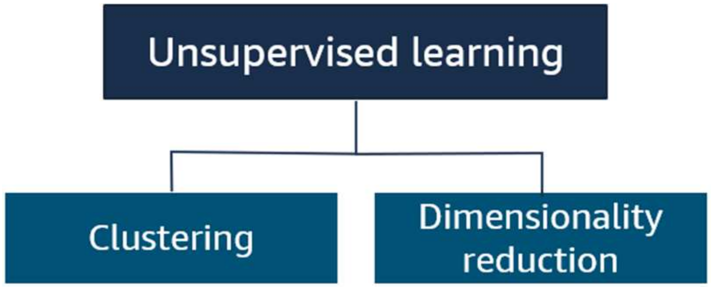
Unsupervised Learning Use Cases

- In unsupervised learning, labels are not provided – you don't know all the variables and patterns.
- The machine has to uncover and create the labels itself.
- The algorithm tries to discover hidden patterns or structures with any prior information or guidance.



Original data

Types of Unsupervised ML



Dimensionality reduction is an unsupervised learning technique used to reduce the number of features or dimensions in a dataset while preserving the most important information or patterns.

Use cases include the following:
- Big data visualization
- Meaningful compression
- Structure discovery
- Feature elicitation

Clustering algorithm groups data into different clusters based on similar features or distances between the data point to better understand the attributes of a specific cluster.

For example, by analyzing customer purchasing habits, an unsupervised algorithm can identify a company as being large or small.

Use cases include the following:
- Customer segmentation
- Targeted marketing
- Recommended systems

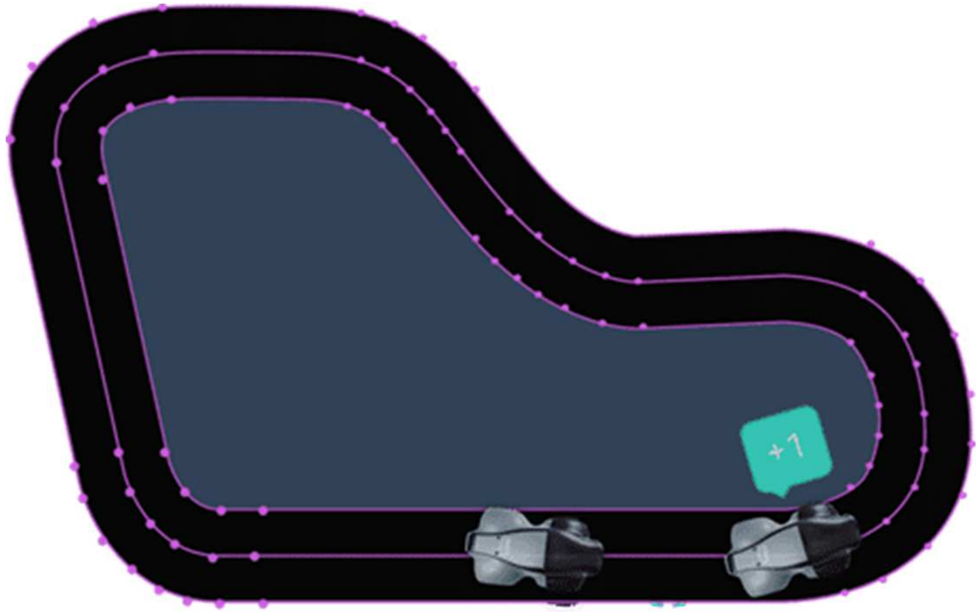
Reinforcement Learning Use Case

In reinforcement learning, an agent continuously learns through trial and error as it interacts in an environment.
Reinforcement learning is broadly useful when the reward of a desired outcome is known but the path to achieving it isn't—and that path requires a lot of trial and error to discover.

AWS DeepRacer and Reinforcement Learning:

- In the AWS DeepRacer simulator, the agent (virtual car) interacts with an environment (virtual racetrack) using actions like throttle and steering to complete the track efficiently.

- The model learns optimal driving behavior through rewards, which incentivize staying on track and reaching the goal quickly.



Generative AI - Capabilities

- Adaptability
- Responsiveness
- Simplicity
- Creativity and Exploration
- Data Efficiency
- Personalization
- Stability

Generative AI - Challenges

Regulatory Violations

- **Risk**

Generative AI models trained on sensitive data might inadvertently generate an output that violates regulations, such as exposing personally identifiable information (PII).

- **Mitigation**

To minimize the risk of privacy violations, implement strict data anonymization and privacy-preserving techniques during model training. To ensure compliance with privacy regulations, conduct thorough audits and assessment of the data used to train the model.

Generative AI - Challenges

Social Risks

- **Risk**

The possibility of unwanted content that might reflect negatively on your organization is a social risk.

- **Mitigation**

Test and evaluate all models before deploying them in production.

Data Security and Privacy Concerns

- **Risk**

The information shared with your model can include personal information and can potentially violate privacy laws.

- **Mitigation**

To protect sensitive data, implement cybersecurity measures, such as encryption and firewalls.

Generative AI - Challenges

- **Toxicity**

Risk

Generative AI models can generate content that is inflammatory, offensive, or inappropriate.

Mitigation

Curate the training data by identifying these phrases in advance and removing them from the training data. This prevents them from being generated as output.

Use guardrail models. These models will detect and filter out unwanted content.

- **Hallucinations**

Risk

The model generates inaccurate responses that are not consistent with the training data. These are called hallucinations.

Mitigation

Teach users that everything must be checked. Foundation models (FM) can't be trusted to verify their own stories are based in reality and on facts. Hallucinations could be further mitigated by checking that content is verified with independent sources. Also, generated content can be marked as unverified to alert the user that verification will be necessary.

Generative AI - Challenges

- **Interpretability**

Risk

Users might misinterpret the model's output, which could lead to incorrect conclusions or decisions.

Mitigation

Use specific domain knowledge for model development and performance by providing key information for data model inputs.

- **Nondeterminism**

Risk

The model might generate different outputs for the same input, which can cause problems in applications where reliability is key.

Mitigation

- Perform tests on the model to identify any sources of nondeterminism. Run the model multiple times and compare the output to ensure consistency.

Factors to Consider When Selecting a Generative AI Model

Factors to consider when selecting an appropriate generative AI model include the following:

- Model types
- Performance requirements
- Capabilities
- Constraints
- Compliance

Models

AI21 labs	• **Jurassic-2 Model** (Text generation, Summarization)
Amazon	• **Amazon Titan** (Text summarization, Classification, Open-ended Q&A, Information extraction, Embeddings)
Anthropic	• **Claude** (Content generation, Text translation)
Stability AI	• **Stable Diffusion** (Improve quality of generated images, Generate photo realistic images from text input)

Models – Performance Requirements

- Performance requirements needs consideration when selecting a generative AI model. These requirements include accuracy, reliability of the output and others.
- Assess the overall performance of the model to evaluate its suitability for a particular task and monitor its performance over time.
- Constraints
- Capabilities
- Compliance
- Cost

Business Metrics for Generative AI

- User Satisfaction
- Average Revenue Per User (ARPU)
- Cross-domain Performance
- Conversion Rate
- Efficiency

Kowledge Check
Q1.

Which artificial intelligence (AI) application is used to automatically extract structured data from various types of documents, such as invoices, contracts, and forms?

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Q2.

A manufacturing company produces a wide range of industrial equipment. Which scenario would be a suitable application of artificial intelligence and machine learning (AI/ML) technology?

A. Implementing a centralized database to store product specifications and manuals.

B. Using computer vision and machine learning algorithms to automate quality control inspections on the production line.

C. Providing employees with training seminars on the latest manufacturing techniques and safety protocols.

D. Outsourcing the manufacturing of certain components to third-party suppliers to reduce costs.

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Q3.

A data engineer is working on a project to predict the sale price of houses based on various features such as square footage, number of bedrooms, location, and age of the property. Which machine learning technique would be appropriate for this task?

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B. Dimensionality reduction
C. Regression
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B. Scalability
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Q5.

A product manager works for a company that uses a generative AI application for content creation. They have noticed that the model often produces different outputs each time it runs with the same input data. What is the challenge this company is encountering?

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